Proton NMR Spectroscopy
Video Series 1

Video 2:
Signals and Integrations

Video Series Goals

Video 1

- Introduction and Theory ☑

Video 2 **(Current)**

- Anatomy of ^{1}H NMR spectra
- Counting Unique Chemical Environments
- Signal Integrations

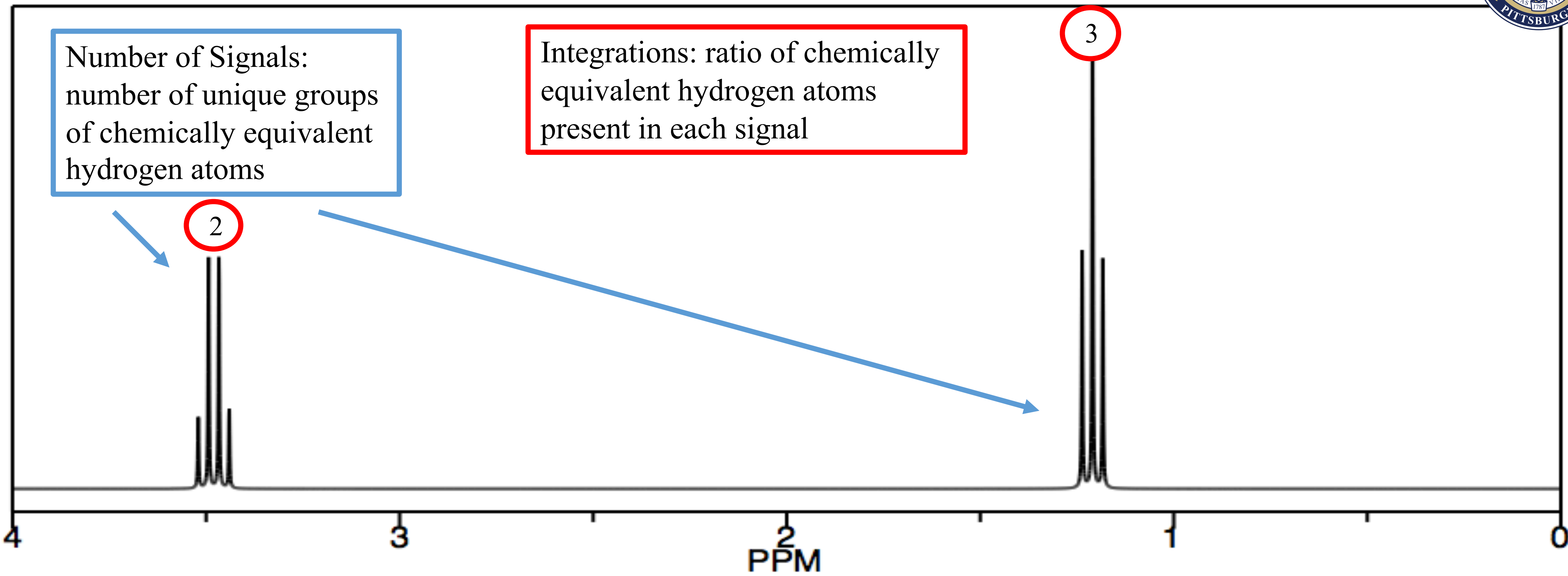
Video 3

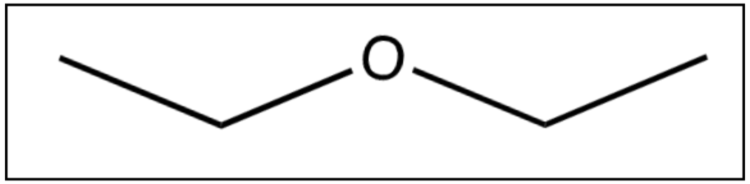
- Chemical Shift

Video 4

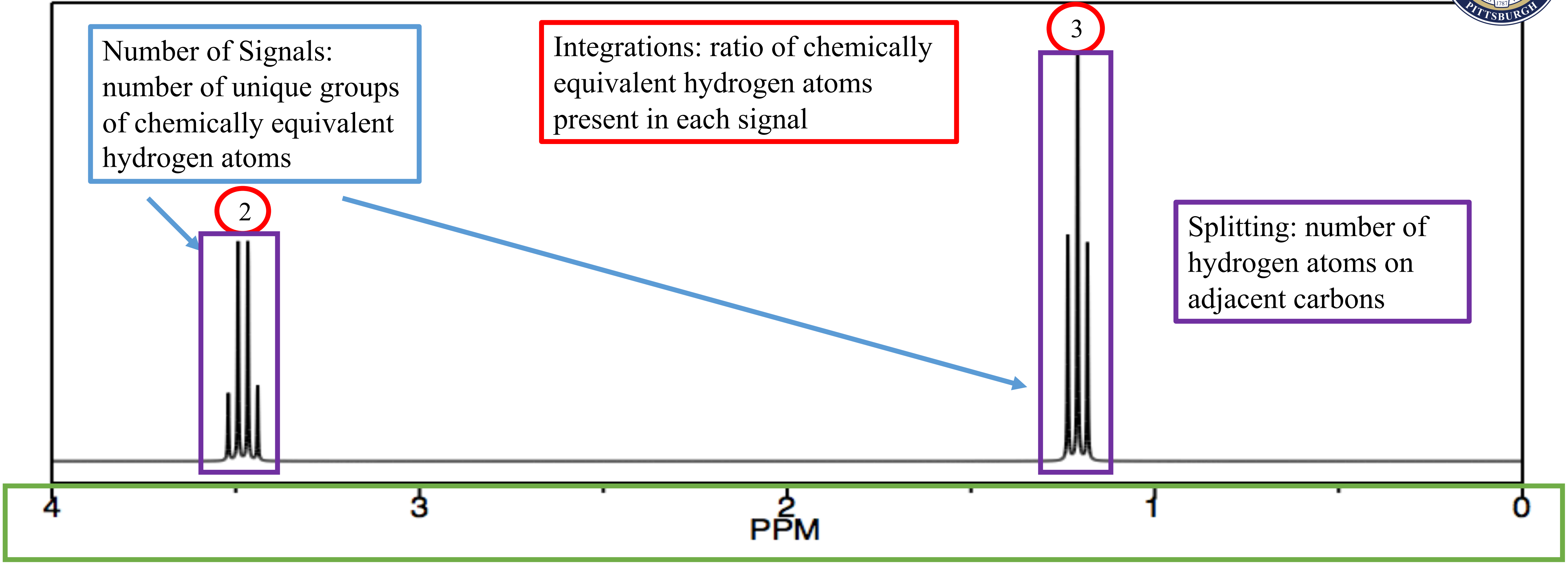
- Splitting Patterns
- Coupling Constants

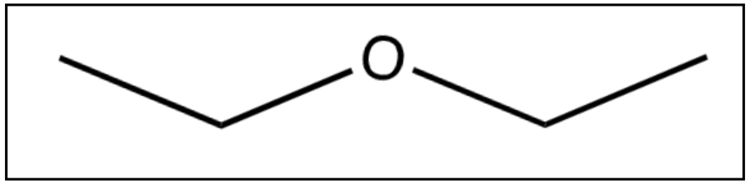
What does a ^{1}H NMR spectrum tell us?





What does a ^{1}H NMR spectrum tell us?

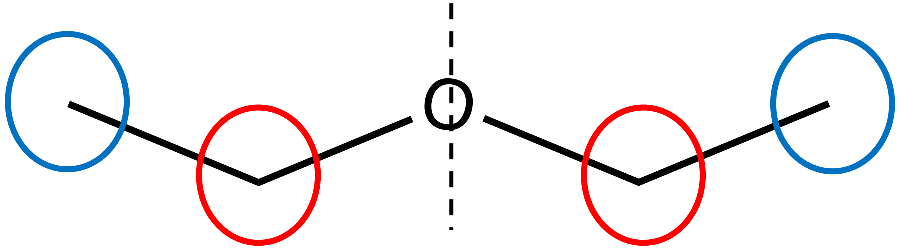


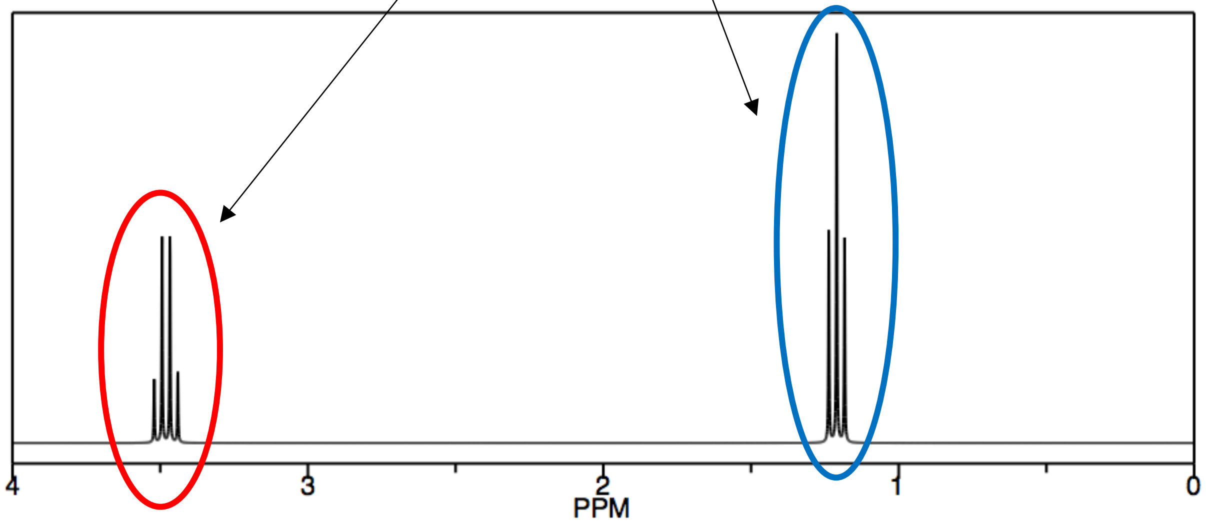


Chemical shift: electronic environment and nearby functional groups

Identifying Unique Chemical Environments

2 clusters of peaks = 2 signals
(2 groups of chemically equivalent H atoms)





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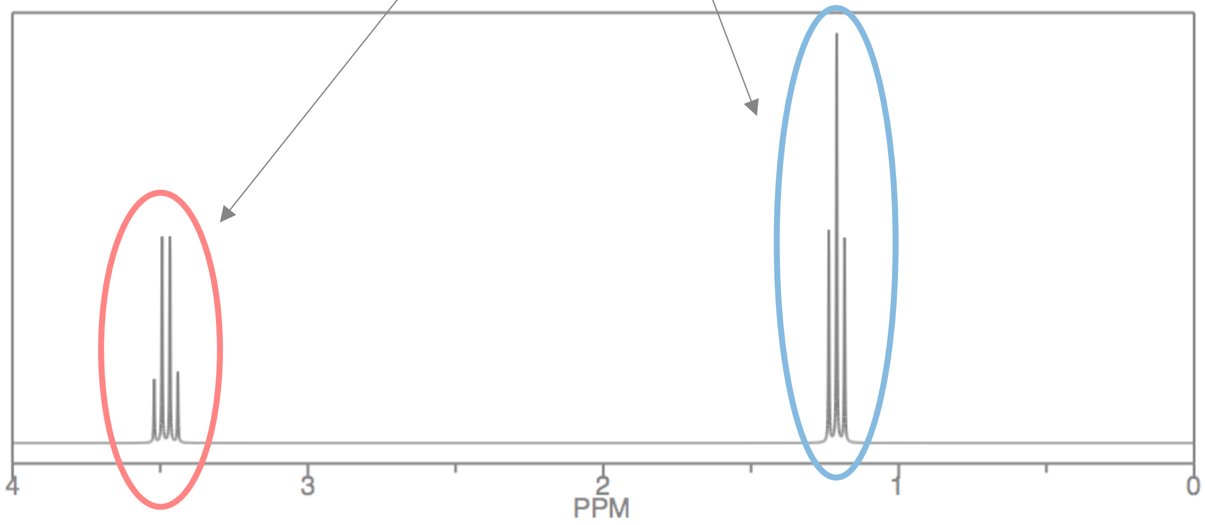
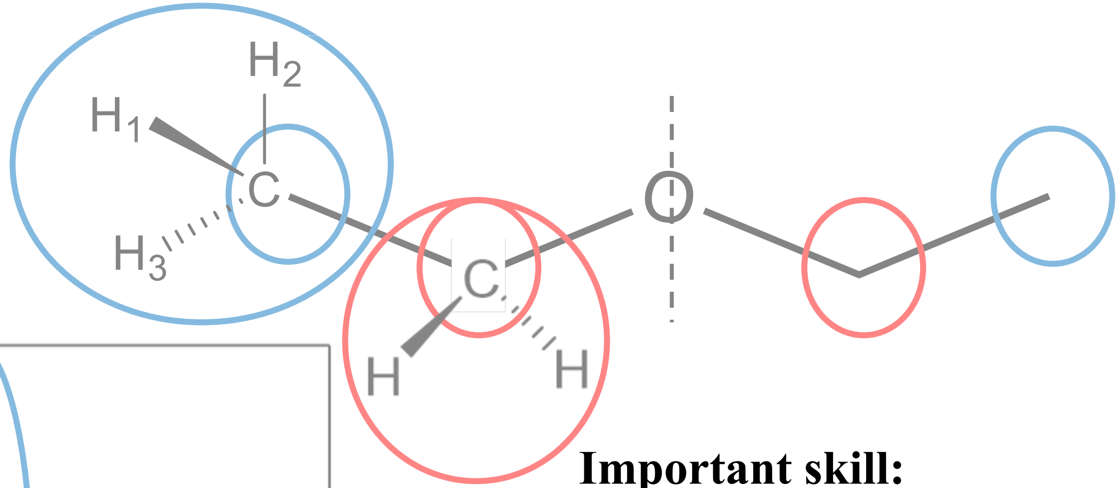


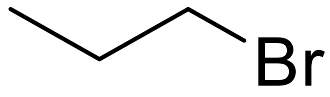


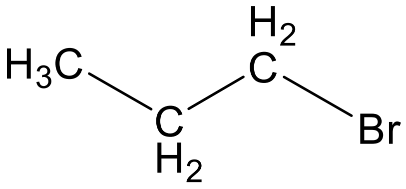
Important skill:
Determine the number of chemically unique protons in chemical compounds

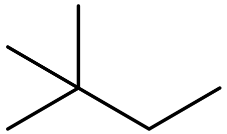
Looking ahead:
Rings, alkenes and compounds with stereocenters can have <u>chemically inequivalent</u> protons <u>on the same carbon atom</u>

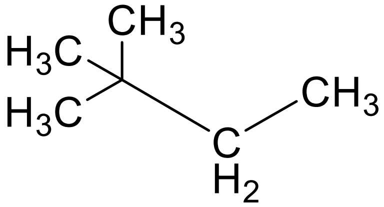
Practice Counting Unique Chemical Environments

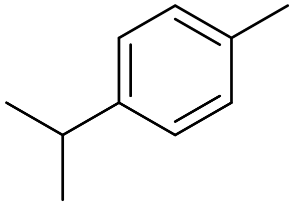
How many sets of chemically equivalent protons are in each of the following compounds?

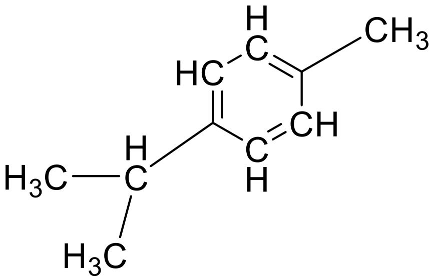






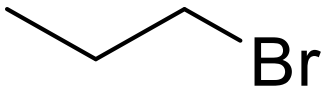


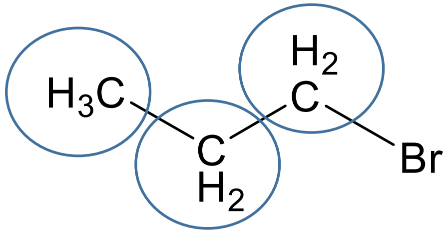


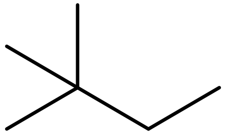


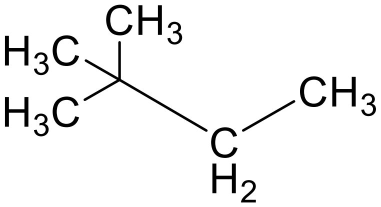
Practice Counting Unique Chemical Environments

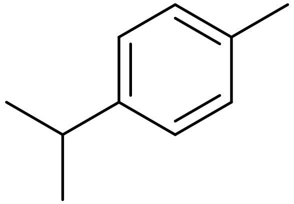
How many sets of chemically equivalent protons are in each of the following compounds?

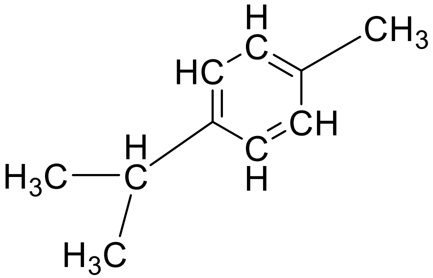




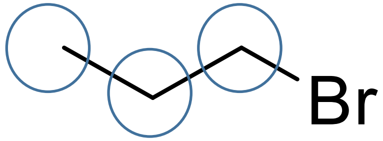








<u>Number of unique chemical environments</u>



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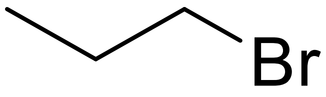
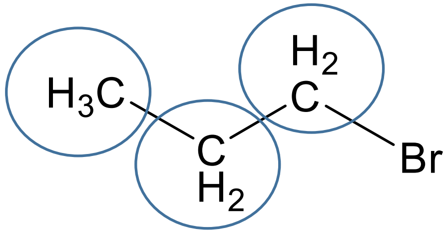
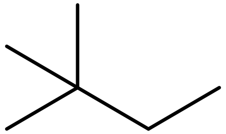
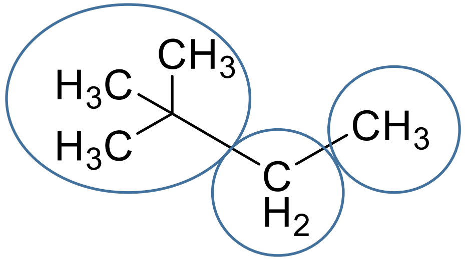
Practice Counting Unique Chemical Environments

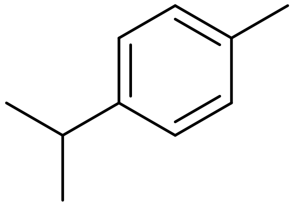
How many sets of chemically equivalent protons are in each of the following compounds?

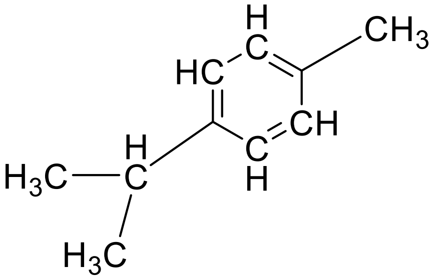




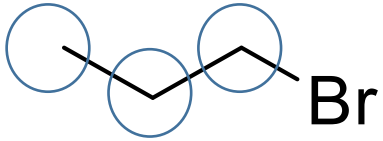








Number of unique chemical environments



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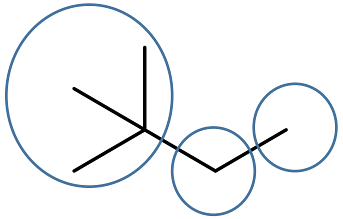

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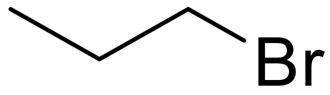
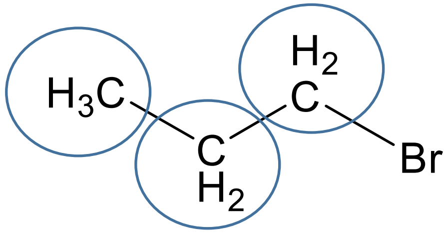
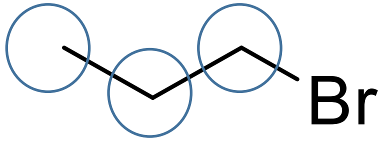
Practice Counting Unique Chemical Environments

How many sets of chemically equivalent protons are in each of the following compounds?

<u>Number of unique chemical environments</u>







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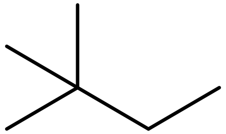

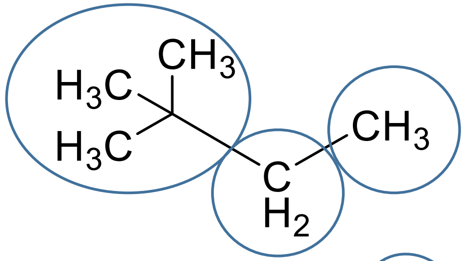

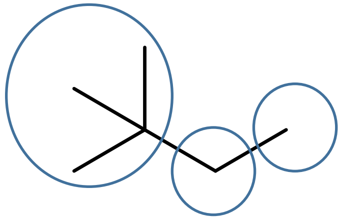

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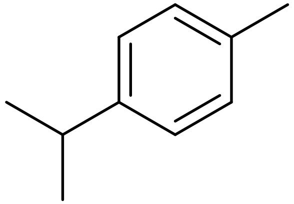

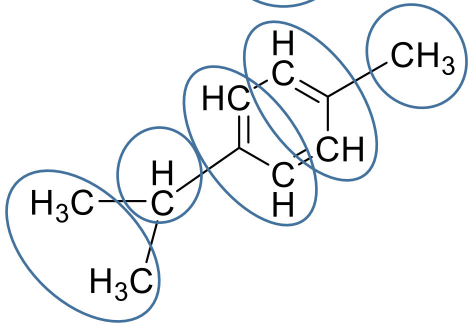

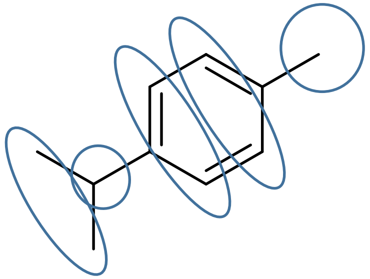

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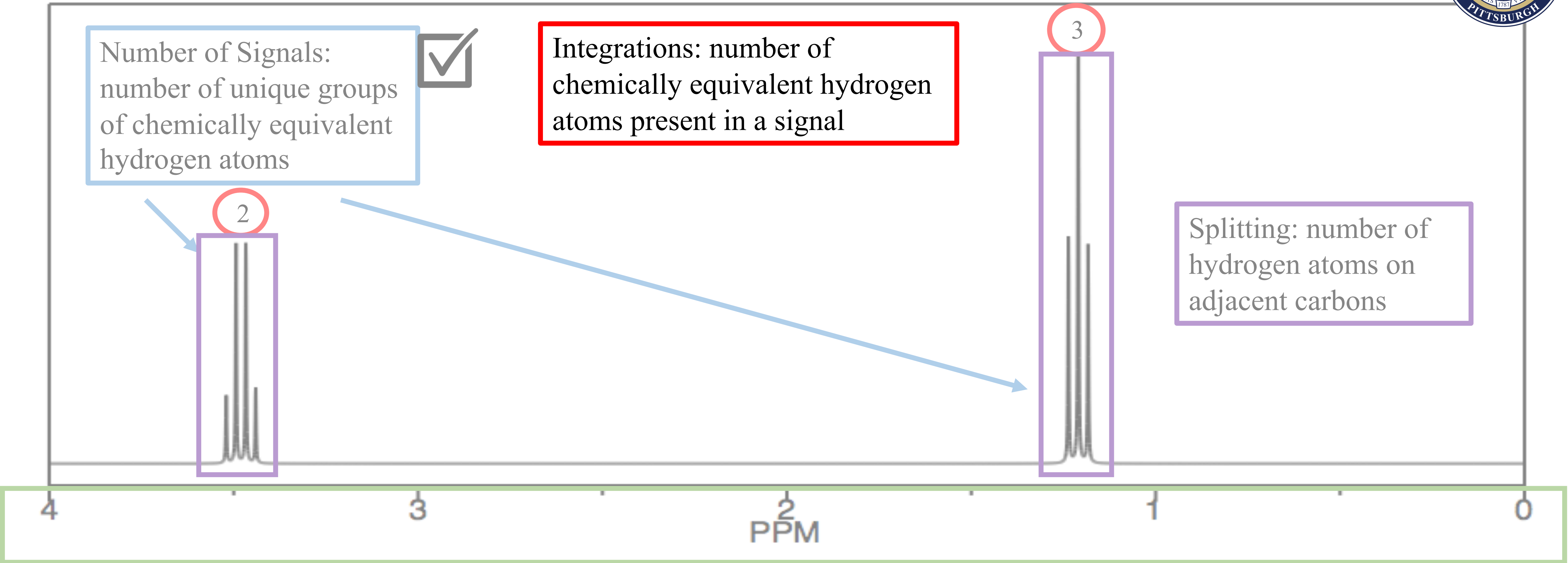
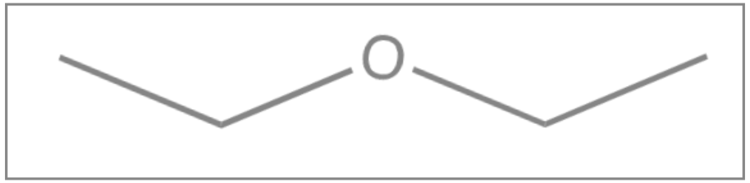
What does a ^{1}H NMR spectrum tell us?





Chemical shift: nearby functional groups and electronic environment

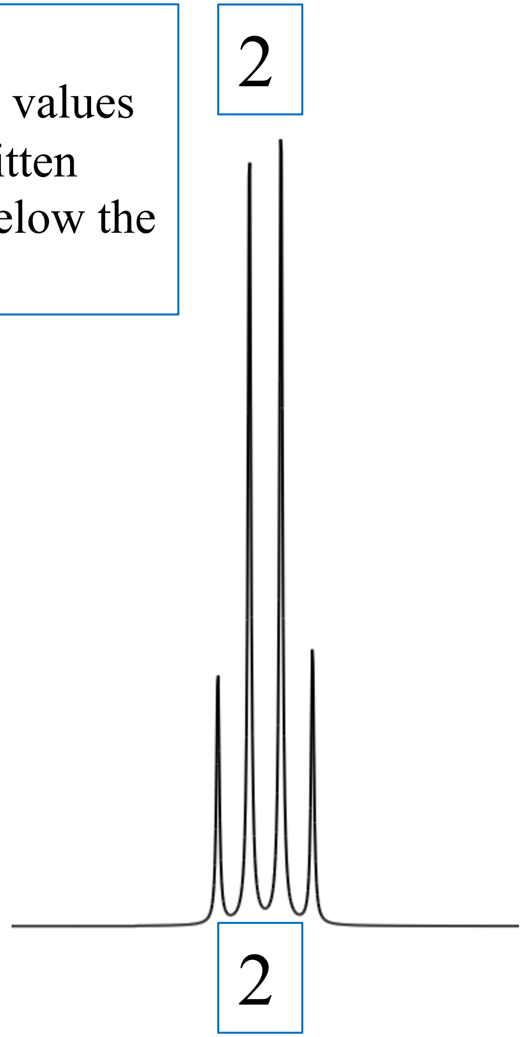
What are signal integrations?

Definition: area under the curve

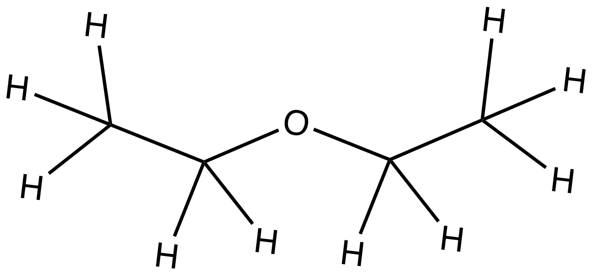
Important Concept: Integration numbers (peak areas) in ^{1}H NMR are proportional to the number of protons that give rise to that signal.

Area under the curve can be calculated by hand, but often is done by computer software.

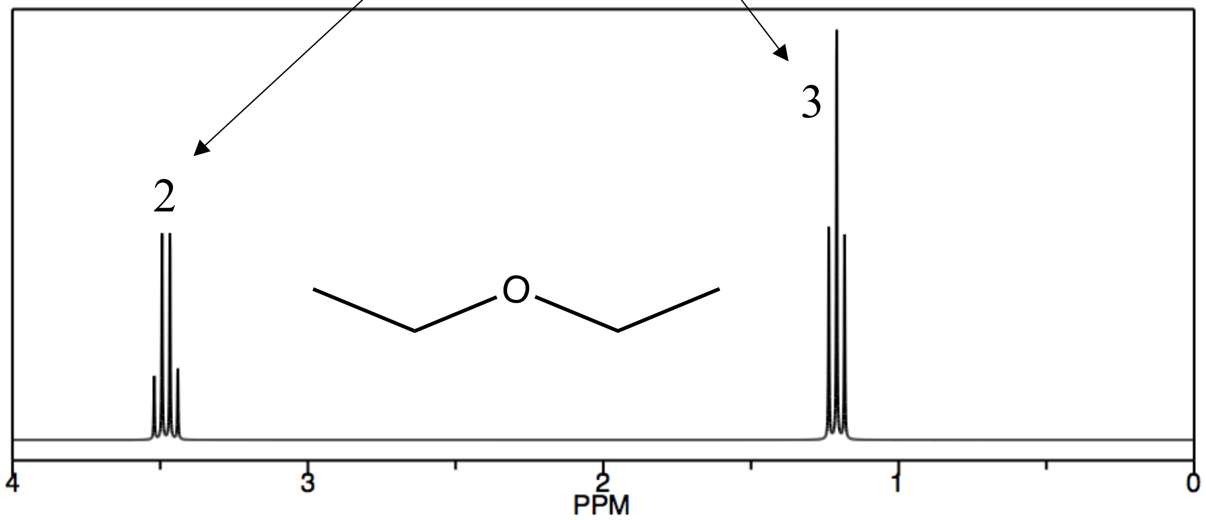
NOTE:
Integration values may be written above or below the signal.



Signal Integrations: Example 1

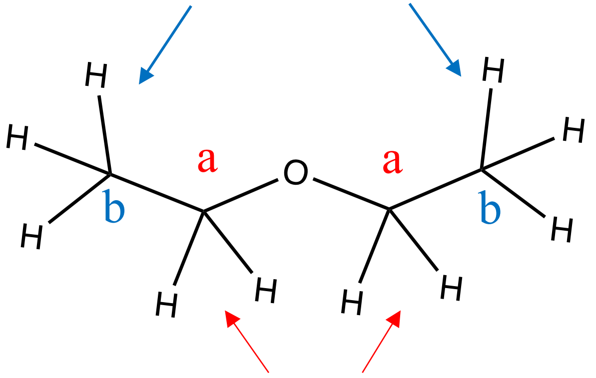


Integration numbers indicate the **ratio** of hydrogen atoms present



Signal Integrations: Example 1

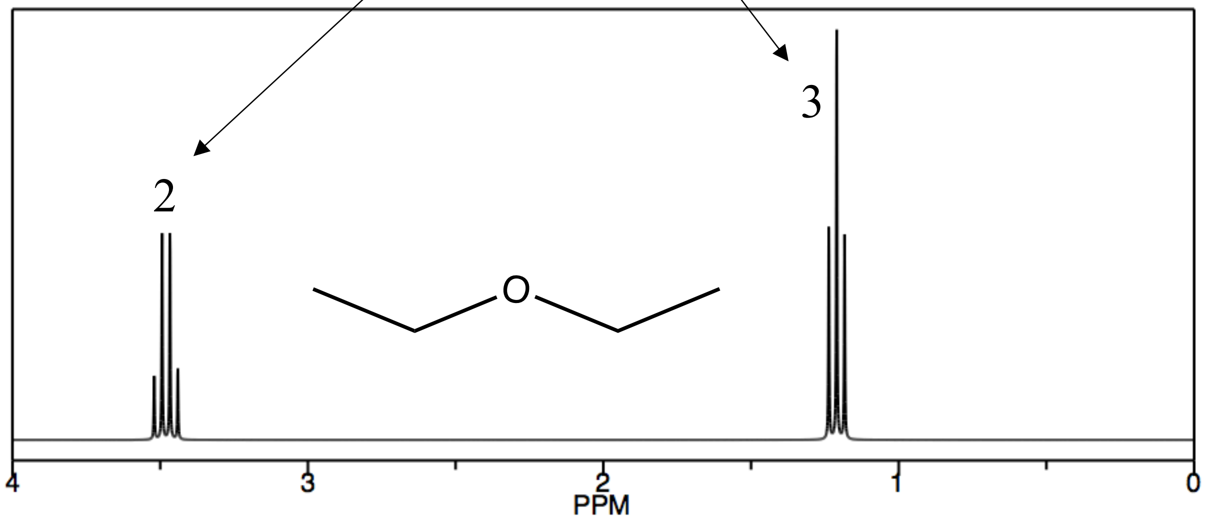
6 chemically equivalent hydrogen atoms



4 chemically equivalent hydrogen atoms

4 : 6 ratio
2 : 3 ratio
1 : 1.5 ratio

Integration numbers indicate the **ratio** of hydrogen atoms present



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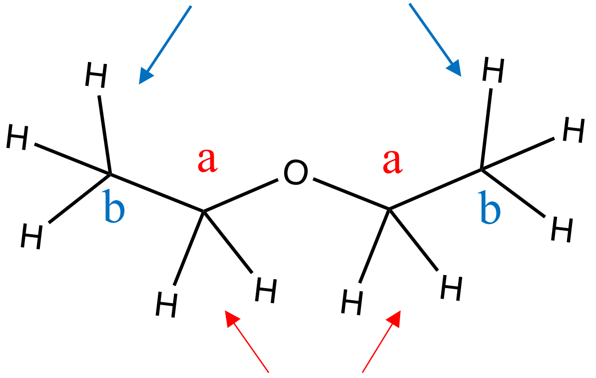

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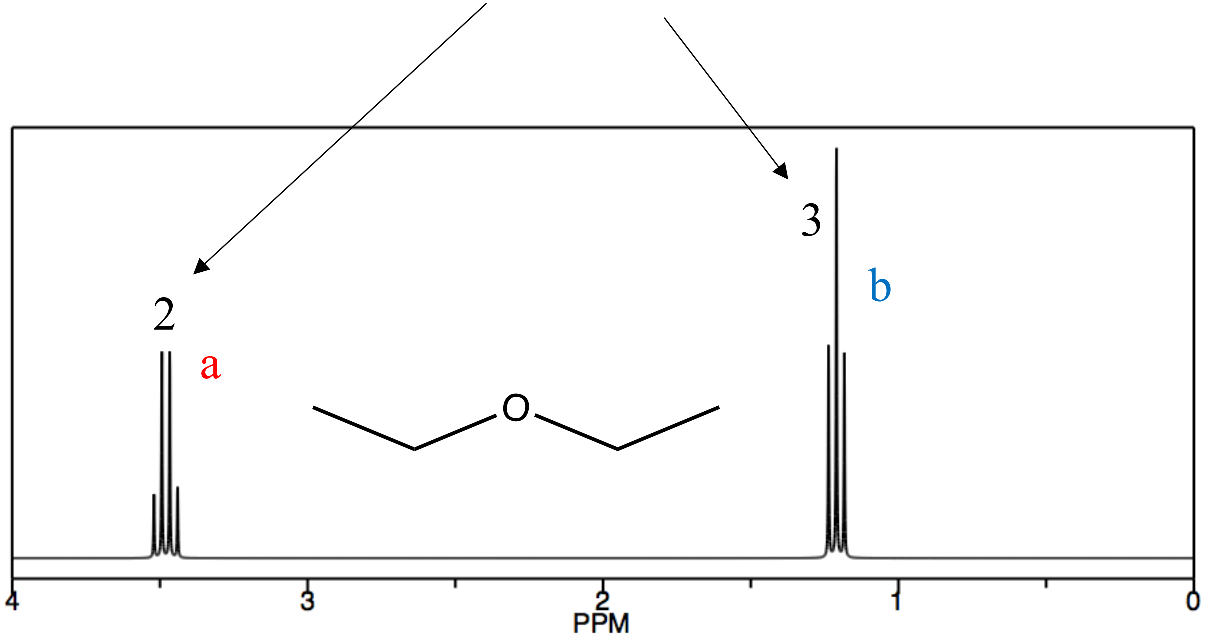
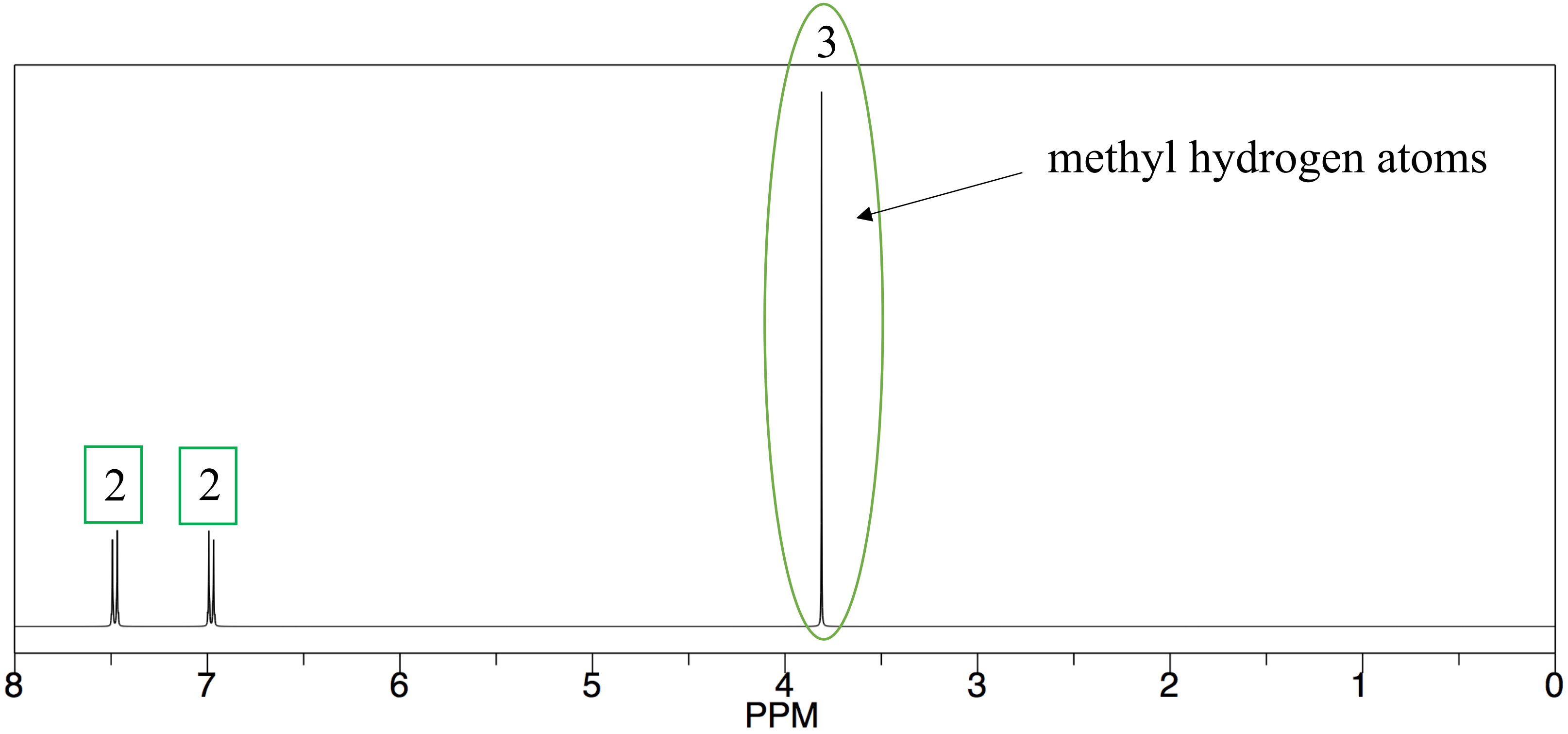
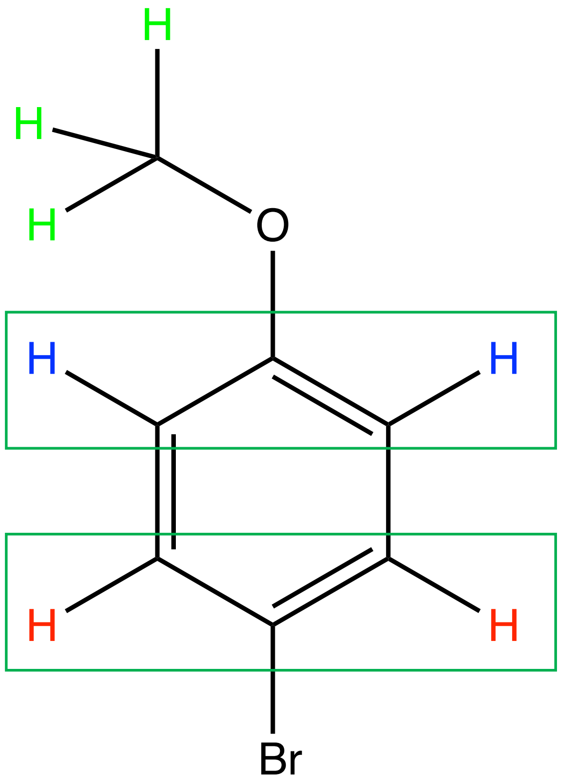


Integrations show a **ratio** and need to be multiplied (by 2 in this case) to represent the absolute number of hydrogen atoms in a signal.

Signal Integrations: Example 2



2 chemically unique types of aromatic hydrogen atoms



Video Summary

The number of signals in a ^{1}H NMR spectrum is related to the number of unique chemical environments in a molecule

- **Determine the number of chemically unique protons in chemical compounds**

Signal peak areas (integrations) are proportional to the number of protons that give rise to that signal

- **One piece of evidence for correct ^{1}H NMR assignments**